

Language

English

Signed in as admin

Log out

AI Job Market 2025–2026

Outputs-first dashboard. New-data processing calls the same isolated offline orchestrator used by the release pipeline.

Data source

Evidence workspace

Baseline release

Upload & process a new dataset

Workflow

- 1. Data Basic Clean
- 2. Data Ready for ML
- 3. AI Job Market Segmentation
- 4. Model Comparison
- 5. Best Model & Importance
- 6. Salary Prediction
- 7. Integrated Market Insight
- 8. Full Pipeline

Active evidence:

FPTCranes-PR32-main

4. Model comparison and temporal validation

Frozen candidates · identical chronological DEV folds · lower MAE is better

Report question

Which frozen candidate performs best on chronological DEV validation, and how stable is that result?

Current takeaway: Random Forest has the lowest mean temporal-validation MAE at \$15,662.

Evidence: 5 candidates across 5 temporal folds · ui-4ee3de59fabafb88

Main limitation: This ranks frozen candidate configurations on DEV folds; it does not replace the saved-model identity or prove a causal fit diagnosis.

Candidates

5

Temporal folds

5

Lowest CV MAE

Random Forest

Best mean CV MAE

\$15,662

Dummy reference

\$54,011

Supplemental evidence ui-4ee3de59fabafb88 . Candidate CV rank does not replace the saved selected model.

> How to read these metrics

> Priority guide: How temporal validation works

Overall comparison Random Forest Gradient Boosting Baseline controls

Question: Which candidate has the lowest temporal-validation error, and how does it compare with training error and the Dummy reference?

Sliding-window Temporal Fold Structure

The development data are divided into six chronological blocks with approximately equal row counts. Each fold uses one block for training and one for validation.

Sliding-window rule: each fold trains on exactly one block and validates on the next block. The Test set remains isolated throughout model selection.

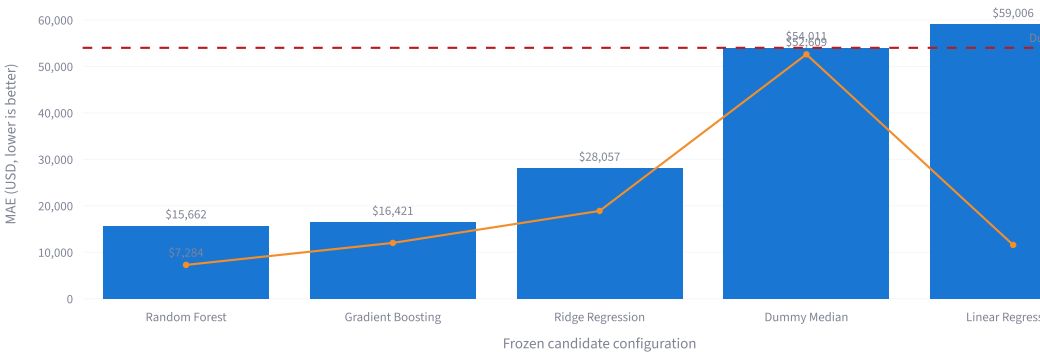
Fold	Block 1	Block 2	Block 3	Block 4	Block 5	Block 6	Test
Fold 1	Train	Validation	—	—	—	—	Test
Fold 2	—	Train	Validation	—	—	—	Test
Fold 3	—	—	Train	Validation	—	—	Test
Fold 4	—	—	—	Train	Validation	—	Test
Fold 5	—	—	—	—	Train	Validation	Test

Recorded sliding-window timeline

Fold	Validation period	Training rows	Validation rows
1	2025-05..2025-08	200	200
2	2025-08..2025-12	200	200
3	2025-12..2026-01	200	200
4	2026-01..2026-02	200	200
5	2026-02	200	201

Adjacent row blocks may share calendar months; row identities—not month labels alone—define each fold.

Train and temporal-validation error



Candidate ranking conclusion

Finding: Random Forest has the lowest mean temporal-validation MAE at \$15,662.

Why it matters: It leads Gradient Boosting by \$759 and is 71.0% lower than the Dummy reference.

Limit: This ranks frozen candidate configurations on DEV folds; it does not replace the saved-model identity or prove a causal fit diagnosis.

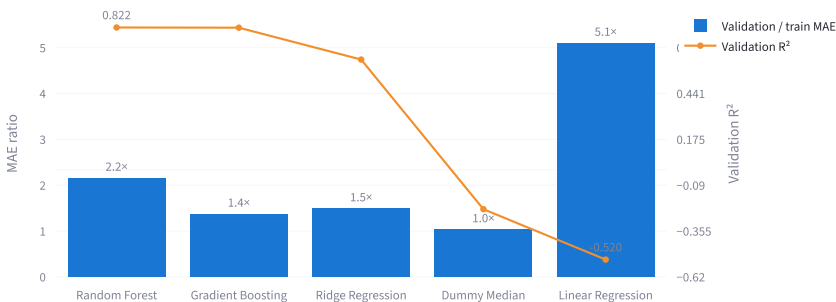
Decision/use: Use this ranking to compare candidate-family evidence, not to auto-promote a model.

Bold = governing result · *Italic = reference/context*

Rank	Model	Decision role	Mean validation MAE	Mean train MAE	Validation R ²
1	Random Forest	Winner	\$15,662	\$7,284	0.822
2	Gradient Boosting	Candidate	\$16,421	\$12,019	0.821
3	Ridge Regression	Candidate	\$28,057	\$18,905	0.637
4	<i>Dummy Median</i>	<i>Reference baseline</i>	<i>\$54,011</i>	<i>\$52,609</i>	<i>-0.228</i>
5	Linear Regression	Candidate	\$59,006	\$11,581	-0.520



Error ratio and explained variance



Model	Rank	Decision role	Validation MAE (USD)	Train MAE (USD)	MAE gap (USD)	Validation R ²	Train R ²	Validation RMSE (USD)	Validation MedAE (USD)
Random Forest	1	Lowest CV MAE	\$15662	\$7284	\$8378	0.822	0.957	\$26821	\$7450
Gradient Boosting	2	Candidate	\$16421	\$12019	\$4402	0.821	0.889	\$26943	\$9450
Ridge Regression	3	Candidate	\$28057	\$18905	\$9152	0.637	0.838	\$38265	\$19980
Dummy Median	4	Reference floor	\$54011	\$52609	\$1401	-0.228	-0.081	\$70447	\$43200
Linear Regression	5	Candidate	\$59006	\$11581	\$47424	-0.520	0.940	\$75094	\$47600

> Additional comparison download

Training logs

> Training log and audit downloads